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FUEL CELL SYSTEM

Abstract

A fuel cell system includes: a fuel cell stack that generates electric power by using a chemical reaction of anode gas and cathode gas; a temperature measurement section that measures temperature of the fuel cell stack; a depressurization section; and an operation control section that controls the fuel cell stack and the depressurization section. The fuel cell stack includes a cathode gas channel in which the cathode gas flows. The depressurization section allows the cathode gas channel to be depressurized. When the operation control section stops the operation of the fuel cell stack, the operation control section controls the depressurization section to cause the depressurization section to depressurize the inside of the cathode gas channel until pressure inside the cathode gas channel falls below the saturated water vapor pressure corresponding to the temperature of the fuel cell stack measured by the temperature measurement section.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION

[0001] This application claims priority to Japanese Patent Application No. 2024-031141 filed on Mar. 1, 2024, incorporated herein by reference in its entirety.

BACKGROUND

1. Technical Field

[0002] The present disclosure relates to a fuel cell system.

2. Description of Related Art

[0003] For example, Japanese Unexamined Patent Application Publication No. 2008-97993 (JP 2008-97993 A) discloses a fuel cell system that removes residual water inside a cathode gas channel by communicating the cathode gas channel with a depressurization section depressurized to pressure lower than the atmospheric pressure when a fuel cell stack is stopped.

SUMMARY

[0004] JP 2008-97993 A studies a fuel cell system that allows residual water inside a fuel cell stack to be efficiently removed so that it is not frozen even when used under a low-temperature environment, but studies nothing about an increase in the durability of the fuel cell system.

[0005] It is possible to implement the present disclosure as the following aspects. [0006] (1) According to an aspect of the present disclosure, a fuel cell system is provided. The fuel cell system includes: a fuel cell stack configured to generate electric power by using a chemical reaction of anode gas and cathode gas; a temperature measurement section configured to measure the temperature of the fuel cell stack; a depressurization section; and an operation control section configured to control the fuel cell stack and the depressurization section. The fuel cell stack includes a cathode gas channel in which the cathode gas flows. The depressurization section is configured to allow the cathode gas channel to be depressurized. The operation control section is configured to, in a case where the operation control section stops the operation of the fuel cell stack, control the depressurization section to cause the depressurization section to depressurize the inside of the cathode gas channel until the pressure of the inside of the cathode gas channel falls below the saturated water vapor pressure corresponding to the temperature of the fuel cell stack measured by the temperature measurement section.

[0007] The fuel cell system according to the aspect makes it possible to decrease the amount of moisture included in a membrane electrode assembly included in the fuel cell stack. It is therefore possible to reduce the deterioration of a cathode catalyst layer included in the membrane electrode assembly and increase the durability of the fuel cell system. [0008] (2) The fuel cell system according to the aspect may include: a pressure measurement section configured to measure the pressure of the inside of the cathode gas channel; and a target pressure setting section configured to set depressurization target pressure that is the pressure of the inside of the cathode gas channel depressurized by the depressurization section. The target pressure setting section may be configured to set the pressure less than the saturated water vapor pressure as the depressurization target pressure. The operation control section may be configured to, in the case where the operation control section stops the operation of the fuel cell stack, control the depressurization section to cause the depressurization section to depressurize the inside of the cathode gas channel until the

pressure measured by the pressure measurement section reaches the depressurization target pressure.

[0009] The fuel cell system according to the aspect makes it possible to control an operation of depressurizing the cathode gas channel by using the pressure of the inside of the cathode gas channel as an indicator. [0010] (3) In the fuel cell system according to the aspect, the target pressure setting section may be configured to set, as the depressurization target pressure, a value of 25% or more of the saturated water vapor pressure and less than 80% of the saturated water vapor pressure.

[0011] The fuel cell system according to the aspect makes it possible to decrease the amount of moisture included in the membrane electrode assembly included in the fuel cell stack and restrain the membrane electrode assembly from being dried too much by the depressurization. [0012] (4) The fuel cell system according to the aspect may further include a stack case in which the fuel cell stack is stored. The depressurization section may be configured to allow the inside of the stack case to be depressurized. The operation control section may be configured to, in the case where the operation control section stops the operation of the fuel cell stack, control the depressurization section to cause the depressurization section to depressurize the inside of the stack case.

[0013] The fuel cell system according to the aspect makes it possible to keep the fuel cell stack warm by insulating the fuel cell stack. It is therefore possible to efficiently depressurize the inside of the cathode gas channel. [0014] (5) The fuel cell system according to the aspect may include: a pressure measurement section configured to measure the pressure of the inside of the cathode gas channel; a membrane resistance measurement section configured to measure membrane resistance that is the electric resistance of a membrane electrode assembly included in the fuel cell stack; and a driving time setting section configured to set the driving time of the depressurization section. The driving time setting section may be configured to set driving scheduled time by using the membrane resistance measured by the membrane resistance measurement section. The driving scheduled time is the driving time necessary to increase a value of the membrane resistance to a predefined target resistance value. The operation control section may be configured to, in the case where the operation control section stops the operation of the fuel cell stack, control the depressurization section to cause the depressurization section to depressurize the inside of the cathode gas channel until the driving scheduled time elapses after the pressure measured by the pressure measurement section falls below the saturated water vapor pressure.

[0015] The fuel cell system according to the aspect makes it possible to control the operation of depressurizing the cathode gas channel by using the driving time of the depressurization section as an indicator.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0016] Features, advantages, and technical and industrial significance of exemplary embodiments of the disclosure will be described below with reference to the accompanying drawings, in which like signs denote like elements, and wherein:

[0017] FIG. 1 is an explanatory diagram illustrating a schematic configuration of a fuel cell system;

[0018] FIG. 2 is a sectional view of a schematic configuration of a fuel cell;

[0019] FIG. 3 is a flowchart of operation stop processing according to a first embodiment;

[0020] FIG. 4 is a state diagram of water;

[0021] FIG. 5 is a diagram illustrating a relationship between pressure of an inside of a cathode gas channel, membrane resistance, and an oxidation amount of catalyst-supporting carbon;

[0022] FIG. 6 is an explanatory diagram illustrating a schematic configuration of a fuel cell system according to a second embodiment;

[0023] FIG. **7** is a flowchart of operation stop processing according to the second embodiment;
[0024] FIG. **8** is an explanatory diagram illustrating a schematic configuration of a fuel cell system according to a third embodiment;
[0025] FIG. **9** is a flowchart of operation stop processing according to the third embodiment;
[0026] FIG. **10** is a flowchart of the operation stop processing according to the third embodiment;
and
[0027] FIG. **11** is a diagram describing driving time of a depressurization section and a change in the membrane resistance.

DETAILED DESCRIPTION OF EMBODIMENTS

A. First Embodiment

[0028] FIG. **1** is an explanatory diagram illustrating a schematic configuration of a fuel cell system **100**. The fuel cell system **100** is mounted, for example, on a fuel cell electric vehicle (FCEV). The fuel cell system **100** includes a fuel cell stack **30**, a temperature measurement section **40**, a membrane resistance measurement section **50**, an anode gas supply and discharge system **60**, a cathode gas supply and discharge system **70**, a load **80**, and a control section **90**.

[0029] The fuel cell stack **30** includes a plurality of fuel cells **10** and a pair of end terminals **21**, **22**. The fuel cells **10** each have a plate shape and are stacked in a stack direction that is the thickness direction of the fuel cells **10**. The fuel cells **10** are electric power generation elements each capable of generating electric power even alone. In the present embodiment, each of the fuel cells **10** is configured as a polymer electrolyte fuel cell. The fuel cell **10** is supplied with anode gas and cathode gas serving as reactant gases and generates electric power by using an electrochemical reaction of the gases. For example, the anode gas is hydrogen and the cathode gas is air. The anode gas is also referred to as fuel gas and the cathode gas is also referred to as oxidant gas.

[0030] FIG. **2** is a sectional view of a schematic configuration of the fuel cell **10**. The fuel cell **10** includes a membrane electrode assembly (MEA) **11**, an anode diffusion layer **15**, a cathode diffusion layer **16**, an anode separator **17**, and a cathode separator **18**.

[0031] The membrane electrode assembly **11** includes an electrolyte membrane **12**, an anode catalyst layer **13**, and a cathode catalyst layer **14**. The electrolyte membrane **12** is an ion-exchange membrane that is formed by using a polymer electrolyte material, for example, a fluorine-based resin including perfluorocarbon sulfonic acid, and has proton conductivity. The electrolyte membrane **12** exhibits favorable electric conductivity under a wet condition. The anode catalyst layer **13** and the cathode catalyst layer **14** are each formed by covering electrically conductive particles, for example, carbon particles, supporting a catalyst, such as platinum or a platinum alloy, with a polyelectrolyte having proton conductivity. The anode catalyst layer **13** is formed on one of the surfaces of the electrolyte membrane **12** and the cathode catalyst layer **14** is formed on the other surface of the electrolyte membrane **12**.

[0032] The anode diffusion layer **15** and the cathode diffusion layer **16** each include, for example, a material having gas permeability and electric conductivity, such as carbon cloth or carbon paper. The anode diffusion layer **15** is formed on the surface of the anode catalyst layer **13** opposite to the surface of the anode catalyst layer **13** in contact with the electrolyte membrane **12**. The cathode diffusion layer **16** is formed on the surface of the cathode catalyst layer **14** opposite to the surface of the cathode catalyst layer **14** in contact with the electrolyte membrane **12**.

[0033] The anode separator **17** and the cathode separator **18** each include a member having gas barrier properties and electric conductivity. The anode separator **17** and the cathode separator **18** are each formed by using, for example, a carbon member, such as dense carbon, obtained by compressing carbon particles to not allow gas to permeate the carbon particles or a metal member, such as stainless steel or titanium steel, subjected to press molding. The anode separator **17** is disposed on the surface of the anode diffusion layer **15** opposite to the surface of the anode diffusion layer **15** in contact with the anode catalyst layer **13**. The anode separator **17** includes an anode gas channel **27** in which anode gas flows. The cathode separator **18** is disposed on the

surface of the cathode diffusion layer **16** opposite to the surface of the cathode diffusion layer **16** in contact with the cathode catalyst layer **14**. The cathode separator **18** includes a cathode gas channel **28** in which cathode gas flows.

[0034] The end terminals **21**, **22** illustrated in FIG. **1** are disposed at both end sections of the fuel cell stack **30** in the stack direction of the fuel cell stack **30**. Specifically, the first end terminal **21** is disposed at one of the end sections of the fuel cell stack **30** and the second end terminal **22** is disposed at the other end section of the fuel cell stack **30**. The first end terminal **21** is provided with openings connected to a cathode gas supply channel **71** and a cathode gas discharge channel **74** described below. The second end terminal **22** is provided with openings connected to an anode gas supply channel **62** and an anode gas discharge channel **64** described below.

[0035] The temperature measurement section **40** measures the temperature of the fuel cell stack **30**. In the example illustrated in FIG. **1**, the temperature measurement section **40** is connected to the second end terminal **22**, but it is sufficient if the temperature measurement section **40** is disposed to allow the temperature of the fuel cell stack **30** to be measured. The temperature measurement section **40** is capable of transmitting the measured temperature to the control section **90**.

[0036] The membrane resistance measurement section **50** measures membrane resistance that is the electric resistance of the membrane electrode assembly **11**. In the example illustrated in FIG. **1**, the membrane resistance measurement section **50** is connected to one of the fuel cells **10**, but the membrane resistance measurement section **50** may be connected to each of the fuel cells **10**. Alternatively, the membrane resistance measurement section **50** may be connected to allow the resistance value of the whole of the fuel cells **10** to be measured. The membrane resistance measurement section **50** is capable of transmitting the measured value of the membrane resistance to the control section **90**.

[0037] The anode gas supply and discharge system **60** supplies anode gas to the fuel cell stack **30** and discharges anode off-gas from the fuel cell stack **30**. The anode off-gas includes anode gas that is not used by the fuel cell stack **30** to generate electric power, moisture, and the like. The anode gas supply and discharge system **60** includes an anode gas tank **61**, the anode gas supply channel **62**, an anode gas inlet valve **63**, the anode gas discharge channel **64**, and an anode gas outlet valve **65**.

[0038] The anode gas tank **61** is a container that stores anode gas to be supplied to the fuel cell stack **30**. The anode gas tank **61** stores, for example, high-pressure hydrogen gas.

[0039] The anode gas supply channel **62** is a channel that leads anode gas to the fuel cell stack **30** from the anode gas tank **61**. An end of the anode gas supply channel **62** is connected to the anode gas tank **61** and the other end of the anode gas supply channel **62** is connected to an opening of the second end terminal **22**.

[0040] The anode gas inlet valve **63** adjusts the amount of anode gas to be supplied from the anode gas tank **61**. The anode gas inlet valve **63** is provided to the anode gas supply channel **62**.

[0041] The anode gas discharge channel **64** is a channel that discharges anode off-gas from the fuel cell stack **30**. An end of the anode gas discharge channel **64** is connected to the opening of the second end terminal **22**. The other end of the anode gas discharge channel **64** is open and anode off-gas is discharged through the opening.

[0042] The anode gas outlet valve **65** opens and closes the anode gas discharge channel **64**. The anode gas outlet valve **65** is provided to the anode gas discharge channel **64**.

[0043] The cathode gas supply and discharge system **70** supplies cathode gas to the fuel cell stack **30** and discharges cathode off-gas from the fuel cell stack **30**. The cathode off-gas includes cathode gas that is not consumed by the fuel cell stack **30** to generate electric power, moisture that is generated in the electric power generation by the fuel cell stack **30**, and the like. The cathode gas supply and discharge system **70** includes the cathode gas supply channel **71**, an air pump **72**, a cathode gas inlet valve **73**, a cathode gas discharge channel **74**, a cathode gas outlet valve **75**, a cathode gas discharge branch channel **76**, a cathode gas branch outlet valve **77**, a depressurization

section **78**, and a pressure measurement section **79**.

[0044] The cathode gas supply channel **71** is a channel that leads cathode gas to the fuel cell stack **30**. An end of the cathode gas supply channel **71** is open. It is possible to take in cathode gas through the opening. The other end of the cathode gas supply channel **71** is connected to an opening of the first end terminal **21**. The cathode gas supply channel **71** is provided with the air pump **72** and the cathode gas inlet valve **73**.

[0045] The air pump **72** compresses cathode gas and supplies the cathode gas to the fuel cell stack **30** via the cathode gas supply channel **71**.

[0046] The cathode gas inlet valve **73** adjusts the amount of cathode gas to be supplied from the air pump **72**.

[0047] The cathode gas discharge channel **74** is a channel that discharges cathode off-gas from the fuel cell stack **30**. An end of the cathode gas discharge channel **74** is connected to an opening of the first end terminal **21**. The other end of the cathode gas discharge channel **74** is open and cathode off-gas is discharged through the opening.

[0048] The cathode gas outlet valve **75** opens and closes the cathode gas discharge channel **74**. The cathode gas outlet valve **75** is provided to the cathode gas discharge channel **74**.

[0049] The cathode gas discharge branch channel **76** is a channel branching from the cathode gas discharge channel **74** between the fuel cell stack **30** and the cathode gas outlet valve **75**. An end section of the cathode gas discharge branch channel **76** is open.

[0050] The cathode gas branch outlet valve **77** opens and closes the cathode gas discharge branch channel **76**. The cathode gas branch outlet valve **77** is provided to the cathode gas discharge branch channel **76**. The cathode gas branch outlet valve **77** is closed while the fuel cell stack **30** is generating electric power.

[0051] The depressurization section **78** is provided to allow the cathode gas channels **28** inside the fuel cell stack **30**, the cathode gas discharge channel **74**, and the cathode gas discharge branch channel **76** to be depressurized. The depressurization section **78** is provided to the cathode gas discharge branch channel **76**. The depressurization section **78** is, for example, a vacuum pump.

[0052] The pressure measurement section **79** measures the pressure of the inside of the cathode gas channel **28**. The pressure measurement section **79** is provided to the cathode gas discharge channel **74** between the fuel cell stack **30** and the cathode gas outlet valve **75**.

[0053] The load **80** is connected to the fuel cell stack **30** via an unillustrated electrical circuit including a DC/DC converter. The load **80** is, for example, a drive motor of a vehicle on which the fuel cell system **100** is mounted. The electrical circuit is provided with a voltage measurement section **81** and an electric current measurement section **82**. The voltage measurement section **81** and the electric current measurement section **82** are capable of measuring an output voltage and an output electric current of the fuel cell stack **30** and transmitting the measurement values to the control section **90**.

[0054] The control section **90** is configured as a computer including a CPU **91** and a memory **92**. The CPU **91** reads out and executes control programs stored in the memory **92** to function as an operation control section **93** and a target pressure setting section **94**.

[0055] The operation control section **93** controls the respective sections of the fuel cell system **100** to cause the fuel cell stack **30** to generate electric power. Specifically, the operation control section **93** controls the valves included in the anode gas supply and discharge system **60** and the cathode gas supply and discharge system **70**, the air pump **72**, and the depressurization section **78** to control the operation of the fuel cell stack **30**.

[0056] The target pressure setting section **94** sets depressurization target pressure that is the pressure of the inside of the cathode gas channel **28** depressurized by the depressurization section **78**.

[0057] FIG. **3** is a flowchart of operation stop processing according to the first embodiment. The operation stop processing is processing of stopping the operation of the fuel cell stack **30**. The

operation stop processing is executed in a case where the control section **90** receives an operation stop signal. The operation stop signal is received, for example, in a case where a switch that starts the vehicle on which the fuel cell system **100** is mounted is turned off.

[0058] In step **S10**, the operation control section **93** stops the air pump **72**.

[0059] In step **S20**, the operation control section **93** closes the anode gas inlet valve **63** and the cathode gas inlet valve **73**.

[0060] In step **S30**, the operation control section **93** closes the anode gas outlet valve **65** and the cathode gas outlet valve **75**. Step **S20** and step **S30** are executed to seal the anode gas channel **27** and the cathode gas channel **28**.

[0061] In step **S40**, the temperature measurement section **40** measures the temperature of the fuel cell stack **30**. The temperature of the fuel cell stack **30** measured here is temperature at the time of stopping the operation of the fuel cell stack **30**.

[0062] In step **S50**, the target pressure setting section **94** sets the pressure less than the saturated water vapor pressure corresponding to the temperature of the fuel cell stack **30** measured in step **S40** as the depressurization target pressure. It is preferable that the target pressure setting section **94** sets a value of 25% or more and less than 80% of the saturated water vapor pressure corresponding to the temperature of the fuel cell stack **30** measured in step **S40** as the depressurization target pressure. It is more preferable that the target pressure setting section **94** sets a value of 25% or more and less than 50% of the saturated water vapor pressure corresponding to the temperature of the fuel cell stack **30** measured in step **S40** as the depressurization target pressure.

[0063] FIG. **4** is a state diagram of water. For example, the saturated water vapor pressure with the fuel cell stack **30** having a temperature of 60° C. is 20 kPa. In a case where the temperature of the fuel cell stack **30** measured in step **S40** is 60° C., the target pressure setting section **94** sets the depressurization target pressure, for example, to 10 kPa, which is the pressure corresponding to 50% of the saturated water vapor pressure with the fuel cell stack **30** having a temperature of 60° C.

[0064] In step **S60** of FIG. **3**, the operation control section **93** brings the depressurization section **78** into operation.

[0065] In step **S70**, the operation control section **93** opens the cathode gas branch outlet valve **77**. This depressurizes the inside of the cathode gas channel **28** by the depressurization section **78**.

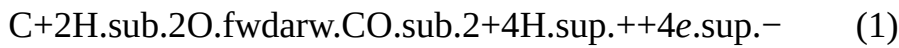
[0066] In step **S80**, the operation control section **93** determines whether or not the pressure measured by the pressure measurement section **79** is less than or equal to the depressurization target pressure. In a case where the pressure is less than or equal to the depressurization target pressure, step **S90** is executed. In a case where the pressure is not less than or equal to the depressurization target pressure, step **S80** is repeated.

[0067] In step **S90**, the operation control section **93** closes the cathode gas branch outlet valve **77**.

[0068] In step **S100**, the operation control section **93** brings the depressurization section **78** out of operation. As described above, the operation stop processing is executed.

[0069] In a case where the operation control section **93** of the fuel cell system **100** according to the first embodiment described above stops an operation of the fuel cell stack **30**, the operation control section **93** controls the depressurization section **78** to cause the depressurization section **78** to depressurize the inside of the cathode gas channel **28** until the pressure of the inside of the cathode gas channel **28** falls below the saturated water vapor pressure corresponding to the temperature of the fuel cell stack **30** measured by the temperature measurement section **40**. Depressurizing the inside of the cathode gas channel **28** causes the moisture inside the cathode gas channel **28** to be vaporized and discharged to the outside of the fuel cell stack **30**. In addition, it is possible to decrease the amount of moisture included in the membrane electrode assembly **11**. The decreased amount of moisture included in the membrane electrode assembly **11** makes an oxidative corrosion reaction of carbon particles supporting a catalyst less likely to occur in the cathode catalyst layer **14** when the fuel cell stack **30** is started next time. The oxidative corrosion reaction is expressed by the

following chemical formula (1).



[0070] The oxidation amount of the carbon particles supporting the catalyst included in the cathode catalyst layer **14** therefore decreases. It is thus possible to reduce the deterioration of the catalyst of the cathode catalyst layer **14** and increase the durability of the fuel cell system **100**. The carbon particles supporting a catalyst will be each referred to as catalyst-supporting carbon below.

[0071] In addition, in a case where the operation control section **93** stops the operation of the fuel cell stack **30**, the operation control section **93** controls the depressurization section **78** to cause the depressurization section **78** to depressurize the inside of the cathode gas channel **28** until the pressure measured by the pressure measurement section **79** reaches the depressurization target pressure in the present embodiment. It is therefore possible to control the operation of depressurizing the cathode gas channel **28** by using the pressure in the cathode gas channel **28** as an indicator.

[0072] FIG. **5** is a diagram illustrating the relationship between the pressure of the inside of the cathode gas channel **28**, the membrane resistance, and the oxidation amount of the catalyst-supporting carbon included in the cathode catalyst layer **14** at the time of the start of the fuel cell stack **30**. FIG. **5** illustrates the membrane resistance and the oxidation amount of catalyst-supporting carbon with the fuel cell stack **30** having a temperature of 60° C. As described above, the saturated water vapor pressure with the fuel cell stack **30** having a temperature of 60° C. is 20 kPa. As illustrated in FIG. **5**, when the inside of the cathode gas channel **28** is depressurized to pressure lower than the saturated water vapor pressure corresponding to the temperature of the fuel cell stack **30**, the amount of moisture included in the membrane electrode assembly **11** decreases. The membrane resistance therefore increases. However, when the membrane electrode assembly **11** is dried and the membrane resistance increases too much, the performance of the fuel cell stack **30** decreases.

[0073] In the present embodiment, the target pressure setting section **94** sets a value of 25% or more and less than 80% of the saturated water vapor pressure corresponding to the temperature of the fuel cell stack **30** as the depressurization target pressure. It is therefore possible to decrease the amount of moisture included in the membrane electrode assembly **11** and restrain the membrane electrode assembly **11** from being dried too much.

B. Second Embodiment

[0074] FIG. **6** is an explanatory diagram illustrating a schematic configuration of a fuel cell system **100b** according to a second embodiment. The fuel cell system **100b** further includes a stack case **110** and a stack case discharge channel **111**.

[0075] The stack case **110** stores the fuel cell stack **30** therein. The stack case **110** includes a material, such as metal, that allows the inside of the stack case **110** to be depressurized.

[0076] The stack case discharge channel **111** is a channel that discharges gas inside the stack case **110**. An end of the stack case discharge channel **111** is connected to the stack case **110**. The other end of the stack case discharge channel **111** is connected to the cathode gas discharge branch channel **76** between the cathode gas branch outlet valve **77** and the depressurization section **78**.

That is, the depressurization section **78** is capable of depressurizing the inside of the stack case **110**.

[0077] FIG. **7** is a flowchart of operation stop processing according to the second embodiment. In the second embodiment, step **S30** to step **S70** are executed in the order of step **S30**, step **S60**, step **S40**, step **S50**, and step **S70**. In the second embodiment, when the depressurization section **78** is brought into operation in step **S60**, the inside of the stack case **110** is depressurized.

[0078] In a case where the operation control section **93** of the fuel cell system **100b** according to the second embodiment described above stops the operation of the fuel cell stack **30**, the operation control section **93** controls the depressurization section **78** to cause the depressurization section **78** to depressurize the inside of the stack case **110**. It is therefore possible to keep the fuel cell stack **30**

warm by insulating the fuel cell stack **30**.

[0079] The following describes, as an example, a case where the temperature of the fuel cell stack **30** is 60° C. at the time of the execution of step **S60** and the target pressure setting section **94** sets the pressure corresponding to 50% of the saturated water vapor pressure corresponding to the temperature of the fuel cell stack **30** measured in step **S40** as depressurization target pressure. In the present embodiment, the fuel cell stack **30** is kept warm by executing step **S60**. In a case where the temperature of the fuel cell stack **30** does not decrease from 60° C., the depressurization target pressure to be set in step **S50** is 10 kPa. However, in a case where the fuel cell system **100** does not include the stack case **110**, the temperature of the fuel cell stack **30** decreases before the depressurization target pressure is calculated. For example, in a case where the temperature of the fuel cell stack **30** decreases to 40° C., the depressurization target pressure is 4 kPa. The motive power of the depressurization section **78** therefore increases more than the motive power of the depressurization section **78** in a case where the fuel cell stack **30** is kept warm. In the present embodiment, the fuel cell stack **30** is kept warm and it is thus possible to reduce the motive power of the depressurization section **78** and decrease the depressurization section **78** in size. In addition, it is possible to efficiently depressurize the inside of the cathode gas channel **28**.

C. Third Embodiment

[0080] FIG. **8** is an explanatory diagram illustrating a schematic configuration of a fuel cell system **100c** according to a third embodiment. A control section **90c** of the fuel cell system **100c** does not include the target pressure setting section **94** and includes a driving time setting section **95**. The configuration of each of the sections of the fuel cell system **100c** other than the control section **90c** is the same as the configuration of the first embodiment.

[0081] The driving time setting section **95** sets the driving time of the depressurization section **78**. The driving time setting section **95** is implemented by the CPU **91** reading out and executing a control program stored in the memory **92**.

[0082] Each of FIG. **9** and FIG. **10** is a flowchart of operation stop processing according to the third embodiment. FIG. **9** and FIG. **10** each denote a step of executing processing similar to processing in FIG. **3** by the same reference sign as the reference sign in FIG. **3** and omit the description.

[0083] In step **S5**, the membrane resistance measurement section **50** measures the membrane resistance.

[0084] In step **S55**, the driving time setting section **95** sets driving scheduled time by using the value of the membrane resistance measured in step **S5**. The driving scheduled time is the driving time of the depressurization section **78** necessary to increase the value of the membrane resistance to a predefined target resistance value. The target resistance value is a value larger than the value of the membrane resistance in a case where the pressure of the inside of the cathode gas channel **28** is the saturated water vapor pressure corresponding to the temperature of the fuel cell stack **30**. It is preferable that the target resistance value is the value of the membrane resistance in a case where the pressure of the inside of the cathode gas channel **28** is 25% or more and less than 80% of the saturated water vapor pressure corresponding to the temperature of the fuel cell stack **30**.

[0085] FIG. **11** is a diagram describing the driving time of the depressurization section **78** and a change in the membrane resistance. FIG. **11** illustrates a change in the membrane resistance under a wet condition by a solid line and a change in the membrane resistance under a dry condition by a dashed line. The wet condition is a condition that the cathode gas channel **28** has liquid water therein when the anode gas channel **27** and the cathode gas channel **28** are sealed. The dry condition is a condition that the cathode gas channel **28** has no liquid water therein when the anode gas channel **27** and the cathode gas channel **28** are sealed. In other words, the wet condition is a condition that the membrane resistance is less than or equal to a reference resistance value when the anode gas channel **27** and the cathode gas channel **28** are sealed. The dry condition is a condition that the membrane resistance is greater than the reference resistance value when the

anode gas channel **27** and the cathode gas channel **28** are sealed. Here, the reference resistance value is the value of the membrane resistance under a condition that the cathode gas channel **28** has liquid water therein, that is, a condition that the electrolyte membrane **12** includes water to the maximum, and a condition that the water vapor pressure of the inside of the cathode gas channel **28** reaches the saturated water vapor pressure corresponding to the temperature of the fuel cell stack **30**. It is determined by using the value of the membrane resistance measured in step **S5** whether the wet condition or the dry condition is applied. It is to be noted that the saturated water vapor pressure depends on the temperature of the fuel cell stack **30** and the relationship between the driving time of the depressurization section **78** and the membrane resistance thus depends on the temperature of the fuel cell stack **30**. FIG. **11** shifts depressurization start time in the case of the dry condition and depressurization start time in the case of the wet condition for the sake of description.

[0086] In the case of the dry condition, depressurization is started at time **T1** and the membrane resistance reaches the target resistance value at time **T2**. The driving time setting section **95** sets the time from time **T1** to time **T2** as driving scheduled time in the case of the dry condition. In the case of the wet condition, depressurization is started at time **T0**, the pressure of the inside of the cathode gas channel **28** falls below the saturated water vapor pressure at time **T1**, and the membrane resistance reaches the target resistance value at time **T3**. The driving time setting section **95** sets the time from time **T1** to time **T3** as driving scheduled time in the case of the wet condition. It is to be noted that the relationship between the driving time of the depressurization section **78** and a change in the membrane resistance illustrated in FIG. **11** is stored in the memory **92** in advance.

[0087] In step **S81** of FIG. **9**, the operation control section **93** determines whether or not the value of the membrane resistance measured by the membrane resistance measurement section **50** is less than or equal to the reference resistance value. In a case where the membrane resistance is less than or equal to the reference resistance value, step **S82** is executed. In a case where the membrane resistance is not less than or equal to the reference resistance value, step **S83** is executed. That is, step **S82** is executed in the case of the wet condition and step **S83** is executed in the case of the dry condition.

[0088] In step **S82**, the operation control section **93** determines whether or not the pressure of the inside of the cathode gas channel **28** measured by the pressure measurement section **79** is less than the saturated water vapor pressure corresponding to the temperature of the fuel cell stack **30** measured by the temperature measurement section **40** in step **S40**. In a case where the pressure is less than the saturated water vapor pressure, step **S83** is executed. In a case where the pressure is not less than the saturated water vapor pressure, step **S82** is repeatedly executed. That is, in the case of the wet condition, the inside of the cathode gas channel **28** is depressurized until the pressure of the inside of the cathode gas channel **28** falls below the saturated water vapor pressure.

[0089] In step **S83**, the operation control section **93** starts to count the driving time of the depressurization section **78**. In other words, the operation control section **93** counts the time elapsed since it is determined in step **S82** that the pressure is less than the saturated water vapor pressure.

[0090] In step **S84**, the operation control section **93** determines whether or not the driving scheduled time has elapsed after it is determined that the pressure is less than the saturated water vapor pressure. In a case where the driving scheduled time has elapsed, step **S90** is executed. In a case where the driving scheduled time has not elapsed, step **S84** is repeatedly executed.

[0091] The driving time setting section **95** of the fuel cell system **100c** according to the third embodiment described above sets the driving scheduled time by using the membrane resistance measured by the membrane resistance measurement section **50**. In a case where the operation control section **93** stops the operation of the fuel cell stack **30**, the operation control section **93** controls the depressurization section **78** to cause the depressurization section **78** to depressurize the inside of the cathode gas channel **28** until the driving scheduled time elapses after the pressure

measured by the pressure measurement section **79** falls below the saturated water vapor pressure. It is therefore possible to control the operation of depressurizing the cathode gas channel **28** by using the driving time of the depressurization section **78** as an indicator.

D. Other Embodiments

[0092] (D-1) In the first embodiment and the second embodiment, the fuel cell systems **100**, **100b** include the membrane resistance measurement sections **50**. In contrast, the fuel cell systems **100**, **100b** do not have to include the membrane resistance measurement sections **50**. [0093] (D-2) In the first embodiment, the fuel cell system **100** includes the target pressure setting section **94**. In contrast, the fuel cell system **100** does not have to include the target pressure setting section **94**. In the case, step **S50** of the operation stop processing illustrated in FIG. **3** is not executed. In addition, in step **S80**, the operation control section **93** determines whether or not the pressure measured by the pressure measurement section **79** is less than the saturated water vapor pressure corresponding to the temperature of the fuel cell stack **30** measured by the temperature measurement section **40**. In a case where the pressure is less than the saturated water vapor pressure, step **S90** may be executed.

[0094] The present disclosure is not limited to the embodiments described above. It is possible to implement the present disclosure in various configurations within the scope that does not depart from the gist of the present disclosure. For example, the technical features in the embodiments corresponding to the technical features in the respective aspects described in the summary of the disclosure can be replaced or combined as appropriate to solve some or all of the problems described above or attain some or all of the effects described above. In addition, the technical features can be deleted as appropriate unless the technical features are described herein as essential.

Claims

1. A fuel cell system comprising: a fuel cell stack configured to generate electric power by using a chemical reaction of anode gas and cathode gas; a temperature measurement section configured to measure temperature of the fuel cell stack; a depressurization section; and an operation control section configured to control the fuel cell stack and the depressurization section, wherein the fuel cell stack includes a cathode gas channel in which the cathode gas flows, the depressurization section is configured to allow the cathode gas channel to be depressurized, and the operation control section is configured to, in a case where the operation control section stops an operation of the fuel cell stack, control the depressurization section to cause the depressurization section to depressurize an inside of the cathode gas channel until pressure of the inside of the cathode gas channel falls below saturated water vapor pressure corresponding to the temperature of the fuel cell stack measured by the temperature measurement section.
2. The fuel cell system according to claim 1, comprising: a pressure measurement section configured to measure the pressure of the inside of the cathode gas channel; and a target pressure setting section configured to set depressurization target pressure that is the pressure of the inside of the cathode gas channel depressurized by the depressurization section, wherein the target pressure setting section is configured to set the pressure less than the saturated water vapor pressure as the depressurization target pressure, and the operation control section is configured to, in the case where the operation control section stops the operation of the fuel cell stack, control the depressurization section to cause the depressurization section to depressurize the inside of the cathode gas channel until the pressure measured by the pressure measurement section reaches the depressurization target pressure.
3. The fuel cell system according to claim 2, wherein the target pressure setting section is configured to set, as the depressurization target pressure, a value of 25% or more of the saturated water vapor pressure and less than 80% of the saturated water vapor pressure.
4. The fuel cell system according to claim 1, further comprising a stack case in which the fuel cell

stack is stored, wherein: the depressurization section is configured to allow an inside of the stack case to be depressurized; and the operation control section is configured to, in the case where the operation control section stops the operation of the fuel cell stack, control the depressurization section to cause the depressurization section to depressurize the inside of the stack case.

5. The fuel cell system according to claim 1, comprising: a pressure measurement section configured to measure the pressure of the inside of the cathode gas channel; a membrane resistance measurement section configured to measure membrane resistance that is electric resistance of a membrane electrode assembly included in the fuel cell stack; and a driving time setting section configured to set driving time of the depressurization section, wherein the driving time setting section is configured to set driving scheduled time by using the membrane resistance measured by the membrane resistance measurement section, the driving scheduled time being the driving time necessary to increase a value of the membrane resistance to a predefined target resistance value, and the operation control section is configured to, in the case where the operation control section stops the operation of the fuel cell stack, control the depressurization section to cause the depressurization section to depressurize the inside of the cathode gas channel until the driving scheduled time elapses after the pressure measured by the pressure measurement section falls below the saturated water vapor pressure.
